CHAPTER 12

Data Warehousing and Data Mining

MCQ Question Bank | Exam-Focused Practice Set

74	7	4
Total Questions	Sections	Options Each

Syllabus Coverage (12.1)

- Concept of Data Warehousing
- Architecture of Data Warehousing
- Various Tools in Data Warehousing
- Tools in Data Mining
- Difference between Data Mining and Normal Query

Contents

Table of Contents

1. Concept of Data Warehousing	Q1–Q10
2. Data Warehousing Architecture	Q11–Q27
3. Data Warehousing Tools	Q28–Q38
4. Data Mining Tools	Q39–Q47
5. Data Mining vs Normal Query	Q48–Q56
6. Important & Tricky Questions	Q57–Q64
7. Mixed OAVS TGT Practice Set	Q65–Q74

Sec. 1 1. Concept of Data Warehousing

Foundational definitions and defining characteristics of a Data Warehouse.

Q1 Concept | Data Warehousing

Which of the following best defines a Data Warehouse?

A A database designed exclusively for real-time transaction processing

B A subject-oriented, integrated, time-variant, and non-volatile collection of data used for decision support

C A single table that stores only the most recent transaction records

D A temporary staging area that is cleared every 24 hours

Correct Answer: B

> **Explanation:** This is Bill Inmon's classic definition, describing the four defining properties of a data warehouse.

Q2 Concept | Subject-Oriented

The term "subject-oriented" in the context of a data warehouse means that data is organized around:

A major business subjects such as sales, customers, and products rather than around applications

B the specific application that originally created it

C the programming language used to store it

D the physical location of the database server

Correct Answer: A

> **Explanation:** Data warehouses group data by business subject to support analysis, unlike operational systems organised by application.

Q3 Concept | Non-Volatile

"Non-volatile" as a property of a data warehouse means:

A data is deleted automatically after every query

B once data is loaded into the warehouse it is only read, not updated or deleted

C data is stored only in volatile RAM memory

D data changes continuously in real time

Correct Answer: B

> **Explanation:** Warehouses store stable, historical data; source-system updates create new records rather than overwriting old ones.

Q4 Concept Time-Variant

The "time-variant" property of a data warehouse means:

A data has no relationship with time at all

B only the current day's data is retained

C data is associated with a specific time period, enabling historical trend analysis

D data is stored for only 24 hours before deletion

Correct Answer: C

Explanation: A warehouse retains historical snapshots so trends can be analysed across months or years, unlike OLTP systems.

Q5 Concept Purpose of DW

A Data Warehouse primarily supports which of the following?

A Network routing between servers

B Decision support and analytical (OLAP-style) processing

C Day-to-day order entry and transaction processing

D Real-time password authentication

Correct Answer: B

Explanation: Warehouses are built for management decision-making and analysis, unlike OLTP systems that handle routine transactions.

Q6 Concept Data Mart

Which term describes a smaller, department-specific subset of a data warehouse?

A Data Mesh

B Data Cube

C Data Lake

D Data Mart

Correct Answer: D

Explanation: A data mart is a scaled-down, subject-specific version of a warehouse built for a single department such as Sales or HR.

Q7 Concept Inmon's Characteristics

Which of these is NOT a defining characteristic of a data warehouse according to Inmon's definition?

A Subject-oriented

B Integrated

C Time-variant

D Volatile (frequently updated in place)

Correct Answer: D

Explanation: Inmon defined a warehouse as non-volatile, not volatile — data is loaded and read, not repeatedly overwritten.

Q8 Concept Integration

The process of combining data from multiple heterogeneous source systems into a consistent format for a data warehouse is called:

A Fragmentation

B Normalization only

C Integration

D Replication

Correct Answer: C

Explanation: Integration ensures naming conventions, units, and formats are made consistent across all sources feeding the warehouse.

Q9 Concept DW vs Data Mart

Which statement correctly distinguishes a Data Warehouse from a Data Mart?

A A data warehouse can serve only one department, while a data mart serves the whole enterprise

B Data marts and data warehouses are simply two names for the exact same thing

C A data warehouse is enterprise-wide in scope, while a data mart is a smaller, subject/department-specific subset

D A data mart always contains more data than the data warehouse it is derived from

Correct Answer: C

Explanation: The warehouse integrates data across the whole organisation; a data mart is a focused, smaller-scope subset for one department.

Q10 Concept Benefits of DW

Which of the following is NOT typically considered a benefit of implementing a data warehouse?

A It consolidates data from multiple sources into a single consistent view

B It eliminates the need for any historical data retention

C It enables historical trend analysis over time

D It supports better, data-driven management decision-making

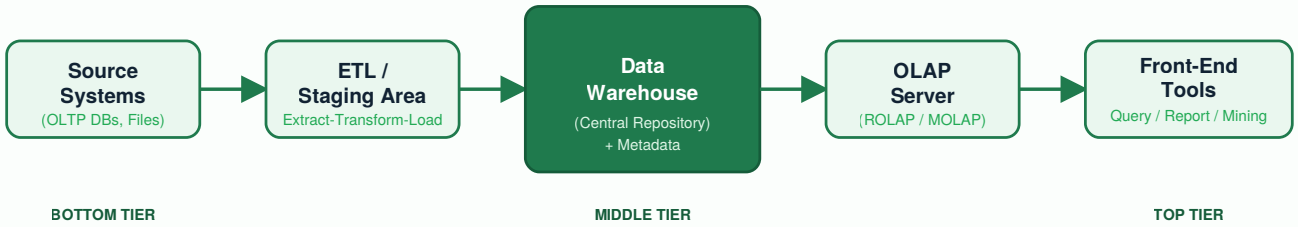
Correct Answer: B

> **Explanation:** A warehouse is valued precisely because it retains historical data for analysis — not because it removes it.

Sec. 2 2. Data Warehousing Architecture

Layers/tiers of a typical data warehouse, ETL, staging, metadata and OLAP components.

Three-Tier Data Warehouse Architecture (Reference Diagram)



Q11 Architecture | Bottom Tier

In a typical three-tier data warehouse architecture, the bottom tier consists of:

A the reporting and query front end

B the OLAP client tools used by analysts

C the data warehouse server and back-end (ETL) tools that extract and load data from operational sources

D the end-user dashboard applications

Correct Answer: C

Explanation: The bottom (data) tier hosts the relational warehouse database along with the back-end ETL tools.

Q12 Architecture | Middle Tier

In the three-tier data warehouse architecture, the middle tier is typically:

A the operational source database

B an OLAP server that provides a multidimensional view of the data

C the end-user reporting screen

D the physical storage disk array

Correct Answer: B

Explanation: The middle tier hosts the OLAP server (ROLAP/MOLAP/HOLAP), presenting data in cube/multidimensional form.

Q13 Architecture Top Tier

The top tier of a three-tier data warehouse architecture is the:

A data mart layer only

B front-end client layer containing query, reporting, and analysis/mining tools

C staging area

D ETL engine

Correct Answer: B

Explanation: The top tier is where end users interact with the warehouse via query, reporting, and analysis tools.

Q14 Architecture ETL

The process of extracting data from sources, cleaning/transforming it, and loading it into a warehouse is known as:

A OLTP

B SQL Injection

C ETL (Extract, Transform, Load)

D Data Replication only

Correct Answer: C

Explanation: ETL is the standard pipeline process used to populate a data warehouse from heterogeneous operational sources.

Q15 Architecture Staging Area

In data warehouse architecture, the "staging area" is primarily used to:

A permanently archive historical warehouse data

B store OLAP cube definitions

C serve live reports directly to end users

D temporarily hold and cleanse raw extracted data before it is loaded into the warehouse

Correct Answer: D

Explanation: The staging area is an intermediate location where data is cleaned and transformed before final loading.

Q16 Architecture Metadata

Metadata in a data warehouse architecture refers to:

A duplicate copies of transactional data

B the physical hard-disk configuration

C data about data — describing the structure, source, and meaning of warehouse data

D encrypted backup files

Correct Answer: C

Explanation: Metadata acts as a directory that helps users and administrators understand what data exists and where it came from.

Q17 Architecture ROLAP

Which architecture stores data in relational tables and uses SQL-based engines at the middle tier to simulate multidimensional analysis?

A HOLAP exclusively

B Flat-file architecture

C ROLAP (Relational OLAP)

D MOLAP only

Correct Answer: C

Explanation: ROLAP performs multidimensional analysis using relational tables and SQL rather than pre-built cubes.

Q18 Architecture Data Marts

In data warehouse architecture, Data Marts are typically positioned:

A as a replacement for the ETL layer

B upstream of the source operational systems

C downstream of the central warehouse, serving specific departments

D completely outside the warehouse architecture

Correct Answer: C

Explanation: Data marts draw a subset of data from the central warehouse (or sources) for a specific business unit.

Q19 Architecture OLAP Engine

Which architectural component allows drill-down and roll-up analysis in a warehouse?

A Firewall
B Load balancer
C OLAP engine / server
D DNS server

Correct Answer: C

Explanation: The OLAP engine provides multidimensional operations such as drill-down, roll-up, slice, and dice.

Q20 Architecture Warehouse Manager

Which component of data warehouse architecture schedules and automates the periodic refresh of warehouse data?

A The warehouse manager / ETL scheduler
B The data mining classifier module
C The end-user reporting dashboard
D The query optimizer of the source OLTP database

Correct Answer: A

Explanation: The warehouse manager component schedules extraction, transformation, and loading jobs to keep the warehouse current.

Q21 Architecture MOLAP

Which OLAP architecture pre-computes and stores data in a multidimensional cube format for very fast query response?

A DOLAP (Desktop OLAP) as the primary server tier
B HOLAP (Hybrid OLAP)
C ROLAP (Relational OLAP)
D MOLAP (Multidimensional OLAP)

Correct Answer: D

Explanation: MOLAP stores data in proprietary multidimensional cubes, giving fast performance but higher storage overhead than ROLAP.

Q22 Architecture HOLAP

Which OLAP architecture combines relational storage for detailed data with cube storage for summarized/aggregated data?

A HOLAP (Hybrid OLAP)
B ROLAP only
C Flat-file storage
D MOLAP only

Correct Answer: A

Explanation: HOLAP blends ROLAP's relational detail storage with MOLAP's fast cube-based aggregation, balancing speed and storage.

Q23 Architecture Two-Tier vs Three-Tier

Compared with a three-tier data warehouse architecture, a two-tier architecture typically:

A adds an extra staging layer not present in the three-tier design
B removes the source/data layer entirely
C merges the OLAP/analysis layer with the client layer, removing the separate middle OLAP-server tier
D is used only for real-time transaction processing, never for analysis

Correct Answer: C

Explanation: A two-tier design has the warehouse database and the client/analysis tools directly interacting, without a dedicated middle OLAP-server tier.

Q24 Architecture Operational Data Store

An Operational Data Store (ODS) in a data warehouse environment is best described as:

A the front-end reporting tool used by analysts
B a type of OLAP cube used only for drill-down
C an integrated, near-real-time staging-like store of current operational data, used before data is loaded into the warehouse
D a permanent archive of decades-old historical data

Correct Answer: C

Explanation: An ODS holds current, lightly-integrated operational data for short-term reporting and as a source feeding the warehouse.

Q25 Architecture Design Approach

Bill Inmon's top-down approach and Ralph Kimball's bottom-up approach to data warehouse design differ mainly in that:

A Inmon builds a single enterprise-wide warehouse first and derives data marts from it, while Kimball builds data marts first and integrates them into a warehouse

B Kimball's approach does not allow any OLAP analysis

C Inmon's approach applies only to data mining, not warehousing

D Inmon's approach uses no ETL process at all, while Kimball's approach requires ETL

Correct Answer: A

Explanation: Inmon = top-down (enterprise warehouse → marts); Kimball = bottom-up (conformed marts → integrated warehouse).

Q26 Architecture Virtual Data Warehouse

A Virtual Data Warehouse is best described as one that:

A provides a logical/virtual view over operational data without physically storing a separate consolidated copy

B is a synonym for a physical three-tier warehouse with an OLAP cube

C always requires the largest possible physical storage infrastructure

D can only be built using MOLAP cube technology

Correct Answer: A

Explanation: A virtual warehouse gives users a consolidated view via queries across source systems in real time, rather than physically storing the data.

Q27 Architecture OLAP Operations

Drill-down, roll-up, slice, dice, and pivot are all examples of:

A Data mining classification algorithms

B Network security protocols

C ETL steps used to load data into the warehouse

D OLAP operations used to navigate and analyse multidimensional data

Correct Answer: D

Explanation: These are the standard OLAP operations that let analysts navigate a data cube at different levels of detail and dimension.

Sec. 3 3. Data Warehousing Tools

Well-known commercial and open-source tools/platforms used to build and manage data warehouses.

Q28 DW Tool ETL Tool

Which of the following is a widely used ETL (Extract-Transform-Load) tool associated with data warehousing?

A Informatica PowerCenter
B Adobe Photoshop
C Wireshark
D WEKA

Correct Answer: A

Explanation: Informatica PowerCenter is a popular commercial ETL tool used to extract, transform, and load data into warehouses.

Q29 DW Tool Microsoft Cloud DW

Microsoft's cloud-based data warehousing service (formerly SQL Data Warehouse) is called:

A Azure DevOps
B Microsoft Access
C Azure Synapse Analytics
D Windows Server

Correct Answer: C

Explanation: Azure Synapse Analytics is Microsoft's cloud data warehousing and big-data analytics platform.

Q30 DW Tool Open-Source ETL

Which of these is a well-known open-source ETL/data-integration tool used in data warehousing?

A AutoCAD
B Tally ERP
C VLC Media Player
D Talend

Correct Answer: D

Explanation: Talend is a popular open-source data integration/ETL tool used to build data warehouse pipelines.

Q31 DW Tool AWS Warehouse

Which cloud-based data warehouse service is offered by Amazon Web Services?

A Amazon Route 53

B Amazon Alexa

C Amazon EC2 Auto Scaling

D Amazon Redshift

Correct Answer: D

Explanation: Amazon Redshift is AWS's fully managed, petabyte-scale cloud data warehouse service.

Q32 DW Tool Google Warehouse

Which tool developed by Google is used as a fully managed, serverless data warehouse?

A Google Chrome

B Google Docs

C Google BigQuery

D Google Analytics dashboard only

Correct Answer: C

Explanation: BigQuery is Google Cloud's serverless data warehouse designed for fast SQL analytics on large datasets.

Q33 DW Tool Oracle Warehouse

Oracle's data warehousing product line, built on its relational database engine, is called:

A Oracle VirtualBox

B Oracle Financial Suite

C Oracle Exadata / Oracle Autonomous Data Warehouse

D Oracle Java Runtime

Correct Answer: C

Explanation: Oracle offers Exadata and the Autonomous Data Warehouse as dedicated platforms for warehousing workloads.

Q34 DW Tool BI/Reporting Tool

Which tool is primarily used for business-intelligence reporting and visualization on top of a data warehouse?

A Git
B XAMPP
C Tableau
D Apache Kafka

Correct Answer: C

Explanation: Tableau is a leading BI/visualization tool used to create dashboards and reports from warehouse data.

Q35 DW Tool SAP Warehouse

SAP's data-warehousing solution built for in-memory analytics is called:

A SAP GUI Scripting
B SAP Fiori Launchpad
C SAP SuccessFactors
D SAP BW/4HANA

Correct Answer: D

Explanation: SAP BW/4HANA is SAP's next-generation data warehouse solution optimised for the in-memory HANA platform.

Q36 DW Tool Open-Source BI/ETL

Which open-source tool offers both ETL (Pentaho Data Integration) and BI/reporting capabilities for data warehousing?

A FileZilla
B Blender
C Pentaho
D Postman

Correct Answer: C

Explanation: Pentaho provides an open-source suite covering data integration (ETL), OLAP analysis, and reporting for warehouses.

Q37 DW Tool Microsoft ETL

Which Microsoft tool is used to build ETL packages that extract, transform, and load data into a SQL Server-based data warehouse?

A Microsoft Teams

B Microsoft Excel Power Query only

C SQL Server Integration Services (SSIS)

D SQL Server Reporting Services (SSRS)

Correct Answer: C

Explanation: SSIS is Microsoft's ETL platform, used to build and automate data integration and transformation workflows for a warehouse.

Q38 DW Tool Visualization Tools

Power BI and QlikView are best classified as tools used to:

A replace the need for an OLAP server entirely

B visualize and build dashboards/reports from data warehouse data

C physically store petabytes of warehouse data

D perform the ETL extraction step from source systems

Correct Answer: B

Explanation: Power BI and QlikView are front-end BI/visualization tools that consume warehouse data; they are not storage or ETL platforms.

Sec. 4 4. Data Mining Tools

Common open-source and commercial platforms used specifically for data mining / pattern discovery.

Q39 DM Tool WEKA

Which of the following is a popular open-source data mining tool developed at the University of Waikato?

A Informatica
B SAP BW
C Redshift
D WEKA

Correct Answer: D

Explanation: WEKA (Waikato Environment for Knowledge Analysis) is a widely used open-source suite of data mining algorithms.

Q40 DM Tool IBM SPSS

Which commercial software from IBM is widely used for advanced data mining and predictive analytics?

A IBM Lotus Notes
B IBM SPSS Modeler
C IBM DB2 Connect
D IBM Cloudant

Correct Answer: B

Explanation: SPSS Modeler is IBM's data mining and predictive-analytics workbench used for building models from data.

Q41 DM Tool RapidMiner

RapidMiner is primarily used as a tool for:

A data mining, machine learning, and predictive-analytics workflows
B website hosting
C video editing
D network firewall configuration

Correct Answer: A

Explanation: RapidMiner provides a visual workflow environment for data preparation, mining, and model building.

Q42 DM Tool Orange

Orange is an open-source tool best known for:

A spreadsheet accounting

B domain name registration

C visual, component-based data mining and machine learning

D PDF editing

Correct Answer: C

Explanation: Orange offers a drag-and-drop visual programming interface for data mining tasks such as classification and clustering.

Q43 DM Tool KNIME

KNIME is a data mining tool best described as a:

A relational database management system

B operating system kernel

C antivirus program

D open-source platform for visual, workflow-based data analytics

Correct Answer: D

Explanation: KNIME (Konstanz Information Miner) lets users build data-mining pipelines using a drag-and-drop node interface.

Q44 DM Tool Python Library

Which Python library is commonly used to implement data mining algorithms such as classification and clustering?

A Flask

B Django

C plain NumPy (without ML routines)

D scikit-learn

Correct Answer: D

Explanation: scikit-learn is a widely used Python library providing ready-made data mining and machine learning algorithms.

Q45 DM Tool Tool Category

Which of the following tools is specifically associated with data mining rather than data warehousing?

A Google BigQuery
B RapidMiner
C Azure Synapse Analytics
D Amazon Redshift

Correct Answer: B

Explanation: RapidMiner is a data mining/analytics tool, whereas Redshift, Synapse, and BigQuery are warehousing platforms.

Q46 DM Tool Common Purpose

Orange, WEKA, RapidMiner, and KNIME are all examples of tools primarily used for:

A storing petabytes of relational transaction data
B extracting patterns and knowledge from data (data mining)
C managing network security
D editing spreadsheets only

Correct Answer: B

Explanation: All four are established data-mining/analytics platforms used to discover patterns, not storage systems.

Q47 DM Tool SAS

Which commercial data mining tool, part of the SAS analytics suite, is widely used for predictive modelling and pattern discovery?

A SAS Print Manager
B SAS Base Installer
C SAS Visual Analytics Login Portal
D SAS Enterprise Miner

Correct Answer: D

Explanation: SAS Enterprise Miner is SAS's dedicated data mining workbench for building predictive and descriptive models.

Sec. 5 5. Data Mining vs Normal Query

Conceptual distinction between direct, condition-based database queries and pattern-discovery via data mining.

Q48 Comparison Normal Query

A normal (SQL) database query typically retrieves:

A specific, already-known facts that exactly match the given conditions

B clusters of similar records without predefined criteria

C previously unknown patterns and hidden relationships in data

D predictive models of future behaviour

Correct Answer: A

Explanation: A normal query answers a precise question the user already has in mind, e.g. "list all customers from Odisha."

Q49 Comparison Data Mining

Data mining differs from a normal database query mainly because data mining:

A requires no data processing at all

B only retrieves rows matching an exact WHERE condition

C discovers previously unknown patterns, trends, and relationships in large datasets

D can only be performed on a single table

Correct Answer: C

Explanation: Data mining goes beyond simple retrieval to uncover hidden knowledge such as patterns and correlations.

Q50 Comparison Identify: Query

Which of the following is an example of a normal query rather than a data-mining task?

A "Segment customers into behavioural groups without predefined labels"

B "Discover which products are frequently bought together"

C "Predict which customers are likely to leave the company"

D "List all employees whose salary is above Rs. 50,000"

Correct Answer: D

Explanation: This is a direct, condition-based retrieval — exactly what SQL queries are designed for.

Q51 Comparison Identify: Mining

Which of the following is an example of a data-mining task rather than a normal query?

A "Count the total number of orders placed today"

B "List products with price greater than Rs. 500"

C "Identify hidden purchasing patterns among customers who buy diapers and beer together"

D "Retrieve the address of customer ID 1023"

Correct Answer: C

Explanation: Discovering an unexpected association (market-basket analysis) is a classic data mining example, not a lookup.

Q52 Comparison Prior Knowledge

A normal query requires the user to know:

A the underlying machine-learning algorithm used

B nothing about the data at all

C only the physical storage location of the server

D exactly what information they are looking for, in advance

Correct Answer: D

Explanation: Queries are precise requests, so the user must already know what specific fact or condition to search for.

Q53 Comparison Exploratory Nature

Data mining is considered a(n) _____ process compared to a normal query, which is _____.

A exploratory process / a direct lookup

B outdated process / modern process

C fully manual process / fully automatic process with no algorithms

D destructive process / non-destructive process

Correct Answer: A

Explanation: Data mining explores data to find unknown insights, while a query directly looks up known, specific information.

Q54 Comparison Key Distinction

Which statement correctly distinguishes data mining from a normal query?

A Data mining cannot use a database as its data source

B A normal query always processes more data than data mining

C Data mining uses statistical/analytical algorithms to find patterns, while a normal query uses exact-match conditions

D A normal query can predict future trends, but data mining cannot

Correct Answer: C

Explanation: Data mining applies algorithms (association, clustering, prediction) whereas queries fetch records via conditions like WHERE.

Q55 Comparison Prediction Example

Predicting a customer's likelihood to default on a loan, based on historical patterns, is an example of:

A a data mining task

B a simple SELECT query

C a normal database JOIN operation

D a database backup operation

Correct Answer: A

Explanation: Prediction from patterns in historical data is a core data-mining application, not a straightforward query.

Q56 Comparison Nature of Output

In terms of output, how does the result of a normal query differ from the result of a data mining process?

A A query returns exact, factual rows matching stated conditions, while data mining returns probabilistic patterns, trends, or predictive insights

B A query always returns a graph, while data mining always returns a single number

C Both always produce identical types of output for the same dataset

D A query's output cannot be stored, while data mining output cannot be viewed

Correct Answer: A

Explanation: Queries give deterministic, exact-match results; data mining gives inferred, often probabilistic knowledge not explicitly stored in the data.

Sec. 6 6. Important & Tricky Questions

Assertion-Reason, Match-the-Following, and Statement-based questions for higher-order exam preparation.

Q57 Assertion-Reason Non-Volatility

Assertion (A): A data warehouse is non-volatile.
Reason (R): Data in a warehouse is updated in real time as transactions occur.

A Both A and R are true, but R is not the correct explanation of A
B A is true, but R is false
C Both A and R are false
D Both A and R are true, and R is the correct explanation of A

Correct Answer: B

Explanation: A warehouse IS non-volatile, but this is because data is loaded periodically, not updated in real time — so R is false.

Q58 Assertion-Reason OLAP Middle Tier

Assertion (A): OLAP servers form the middle tier of a typical three-tier data warehouse architecture.
Reason (R): OLAP servers provide a multidimensional view of data for analysis.

A A is true, but R is false
B Both A and R are false
C A is false, but R is true
D Both A and R are true, and R is the correct explanation of A

Correct Answer: D

Explanation: The OLAP server sits in the middle tier precisely because it presents data multidimensionally for analysis.

Q59 Assertion-Reason Automation in Mining

Assertion (A): Data mining always requires the end user to manually write exact-match queries.
Reason (R): Data mining algorithms automatically search for hidden patterns in data.

A Both A and R are true, and R explains A
B Both A and R are false
C A is false, but R is true
D Both A and R are true, but R does not explain A

Correct Answer: C

Explanation: Data mining is largely automated by algorithms discovering patterns; it does not need manual exact-match queries.

Q60 Match-the-Following Tools & Concepts

Match Column A with Column B:
Column A: 1. Redshift 2. WEKA 3. Data Mart 4. ETL
Column B: (i) Data mining tool (ii) Department-specific warehouse subset (iii) AWS data warehouse service (iv) Extract-Transform-Load process

A 1-ii, 2-iv, 3-i, 4-iii

B 1-iii, 2-i, 3-ii, 4-iv

C 1-i, 2-iii, 3-iv, 4-ii

D 1-iv, 2-ii, 3-iii, 4-i

Correct Answer: B

Explanation: Redshift → AWS warehouse; WEKA → data mining tool; Data Mart → departmental subset; ETL → data pipeline process.

Q61 Match-the-Following Tools & Properties

Match Column A with Column B:
Column A: 1. BigQuery 2. RapidMiner 3. Subject-oriented 4. Non-volatile
Column B: (i) Data organised around business subjects (ii) Data is loaded, not updated (iii) Google's serverless warehouse (iv) Data mining/predictive analytics tool

A 1-i, 2-ii, 3-iii, 4-iv

B 1-iv, 2-iii, 3-ii, 4-i

C 1-iii, 2-iv, 3-i, 4-ii

D 1-ii, 2-i, 3-iv, 4-iii

Correct Answer: C

Explanation: BigQuery → Google warehouse; RapidMiner → mining tool; Subject-oriented → business-subject grouping; Non-volatile → read-only after load.

Q62 Statement-Based Integration vs Transactions

Statement I: A data warehouse integrates data from multiple heterogeneous sources.
Statement II: A data warehouse is optimized primarily for fast, high-volume transaction inserts.

A Statement I is true, Statement II is false

B Both statements are false

C Both statements are true

D Statement I is false, Statement II is true

Correct Answer: A

Explanation: Warehouses integrate diverse sources but are optimised for analytical reads (OLAP), not transactional inserts (OLTP's role).

Q63 Statement-Based Staging Area Role

Statement I: The staging area cleans and temporarily holds data before loading into the warehouse.
Statement II: The staging area is the final destination for permanent historical storage.

A Both statements are true
B Both statements are false
C Statement I is true, Statement II is false
D Statement I is false, Statement II is true

Correct Answer: C

Explanation: The staging area is a temporary intermediate step; the warehouse itself is the permanent historical store.

Q64 Statement-Based Query vs Mining (Tricky)

Statement I: Every SQL query performed on a warehouse is considered data mining.
Statement II: Data mining specifically involves discovering non-obvious patterns using statistical/ algorithmic techniques.

A Statement I is true, Statement II is false
B Statement I is false, Statement II is true
C Both statements are true
D Both statements are false

Correct Answer: B

Explanation: Not all queries are data mining — only analyses that uncover hidden, non-obvious patterns qualify as data mining.

Sec. 7 7. Mixed OAVS TGT Practice Set

Scenario-based and mixed-recall questions drawing across the full chapter for final practice.

Q65 Scenario Serverless Warehouse

Which of the following tools would you use to build a fully managed, serverless cloud data warehouse without provisioning servers?

A Orange
B Google BigQuery
C KNIME
D WEKA

Correct Answer: B

Explanation: BigQuery is serverless — users run queries without managing infrastructure, unlike the listed mining tools.

Q66 Architecture Analyst's Layer

Which layer of the data warehouse architecture would a business analyst directly interact with to generate a sales report?

A Top tier (front-end query/reporting tools)
B ETL scheduler
C Bottom tier (source databases)
D Staging area

Correct Answer: A

Explanation: Analysts use front-end tools in the top tier — reporting, query, and analysis applications — to generate reports.

Q67 Scenario Association Analysis

A retail company wants to find out "which items are commonly bought together" without knowing this in advance. This scenario is best solved using:

A A simple SELECT query with a WHERE clause
B A database index rebuild
C A firewall configuration change
D Data mining (association analysis)

Correct Answer: D

Explanation: Discovering unknown item associations from transactions is a classic data mining (market-basket) application.

Q68 DW/DM Tool Category Check

Which of the following would NOT typically be classified as a data warehousing tool?

A Google BigQuery
B Azure Synapse Analytics
C Amazon Redshift
D WEKA

Correct Answer: D

Explanation: WEKA is a data mining tool, while Redshift, Synapse, and BigQuery are data warehousing platforms.

Q69 Concept Why Time-Stamped Data

Which of the following best explains why data warehouses store historical, time-stamped data?

A To reduce the total storage space required
B To eliminate the need for backups
C To allow trend analysis and comparison across different time periods
D To speed up single-row transaction updates

Correct Answer: C

Explanation: Time-variant storage lets analysts compare performance across months, quarters, or years.

Q70 Concept DW vs Operational DB

Which of these best distinguishes a data warehouse from an operational database?

A A data warehouse is designed for analysis and decision support; an operational database is designed for daily transaction processing
B An operational database always stores more data than a warehouse
C Both serve exactly the same purpose
D A data warehouse cannot store any historical data

Correct Answer: A

Explanation: This is the fundamental OLAP (analysis) vs OLTP (transaction) distinction tested frequently in exams.

Q71 Scenario Predictive Pattern

A school wants to identify students at risk of failing, based on patterns in past attendance and marks, without a specific student in mind. This is an example of:

A A simple attendance query for one student

B A network configuration task

C A backup operation

D Data mining (predictive pattern discovery)

Correct Answer: D

Explanation: Identifying an unknown at-risk pattern across many students is a data mining/predictive analytics task.

Q72 Concept Data Marts (True Statement)

Which of these is TRUE regarding Data Marts?

A They replace the need for ETL processes entirely

B They always contain more data than the central warehouse

C They are identical in scope to the entire data warehouse

D They contain a subset of warehouse data focused on a specific department or subject

Correct Answer: D

Explanation: Data marts are smaller, targeted extracts of the warehouse built for specific departments like HR or Sales.

Q73 DW/DM Tool Correct Pairing

Which tool combination correctly matches a cloud data warehouse WITH a data mining platform?

A Orange (warehouse) + BigQuery (mining)

B KNIME (warehouse) + Synapse (mining)

C Amazon Redshift (warehouse) + RapidMiner (mining)

D WEKA (warehouse) + Redshift (mining)

Correct Answer: C

Explanation: Redshift is a warehousing service; RapidMiner is a mining tool — correctly matching category to function.

Q74 Comparison One-Line Distinction

Which of the following best captures the key difference between a normal query and data mining, in one line?

A A query answers a known question; data mining discovers unknown patterns

B Data mining never uses a database as a source

C A query can predict the future while data mining cannot

D A query is always slower than data mining

Correct Answer: A

Explanation: This concise contrast is the most exam-relevant way to remember the distinction between the two.